

Comparative Evaluation of Statistical and Deep Learning Methods for Hyperspectral Anomaly Detection

Independent Research Report

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Abstract

Hyperspectral imagery provides detailed spectral measurements that can support the identification of materials and objects that are difficult to distinguish using conventional imagery. However, hyperspectral anomaly detection remains challenging because of high dimensionality, strong correlations between spectral bands, complex background distributions, and severe class imbalance. This independent study investigates statistical and deep representation-learning approaches for unsupervised anomaly detection using the AVIRIS-1 San Diego hyperspectral dataset.

The analyzed scene contains 10,000 pixels represented by 189 spectral bands, including 9,936 background pixels and 64 aircraft anomaly pixels. Six approaches were investigated: a regularized Reed–Xiaoli (RX) detector, a dense spectral autoencoder, a one-dimensional convolutional autoencoder, a three-dimensional spectral-spatial autoencoder, a spectral variational autoencoder, and a PatchCore-style nearest-neighbor detector based on learned spectral embeddings.

The RX detector achieved a ROC-AUC of 0.9753, demonstrating strong anomaly-ranking capability, but its precision–recall performance and threshold-based detection were substantially weaker under severe class imbalance. Among the representation-learning methods, the PatchCore-style approach produced a ROC-AUC of 0.9346 and the highest observed PR-AUC of 0.1441, together with a precision of 0.2100, recall of 0.3281, and F1-score of 0.2561 at the evaluated operating threshold. In contrast, several reconstruction-based autoencoders achieved relatively high ROC-AUC values while failing to identify anomalies effectively after percentile-based thresholding.

The results demonstrate that strong ranking performance does not necessarily translate into reliable binary anomaly detection in highly imbalanced hyperspectral data. They also highlight the importance of evaluating hyperspectral anomaly detectors using precision–recall measures, false-alarm behavior, spatial anomaly maps, and threshold sensitivity in addition to ROC-AUC.

Keywords: Hyperspectral imaging, anomaly detection, remote sensing, AVIRIS, autoencoder, variational autoencoder, deep learning, representation learning, RX detector, PatchCore

1 Introduction

Hyperspectral remote sensing records information across a large number of contiguous spectral bands. Unlike conventional RGB imagery, in which each pixel contains only a small number of color measurements, a hyperspectral pixel can be represented by a high-dimensional spectral signature. These signatures provide information that can help distinguish materials and objects with similar visual appearance but different spectral properties.

One important application of hyperspectral imagery is anomaly detection. The objective is to identify pixels or spatial regions whose spectral behavior differs substantially from the dominant background. In many practical settings, labels describing anomalous objects may be unavailable during model

development. This motivates unsupervised anomaly-detection approaches that learn the structure of the background or the overall data distribution and assign larger anomaly scores to unusual observations.

Hyperspectral anomaly detection presents several computational challenges. First, the number of spectral bands is large relative to the number of observations available in many scenes. Second, neighboring spectral bands are often highly correlated. Third, anomalous pixels may represent only a very small fraction of the image. Finally, complex urban or natural backgrounds may contain multiple spectral distributions, making it difficult to describe the background using a single statistical model.

This study investigates these challenges through a comparative analysis of statistical and representation-learning approaches using AVIRIS hyperspectral imagery. Rather than evaluating only one algorithm, the project progressively examines models with different assumptions about spectral and spatial structure.

The following approaches are investigated:

1. Regularized Reed–Xiaoli anomaly detection;
2. Dense spectral autoencoder;
3. One-dimensional convolutional autoencoder;
4. Three-dimensional spectral-spatial autoencoder;
5. Spectral variational autoencoder;
6. PatchCore-style anomaly detection using learned spectral embeddings.

The primary research question is:

How do statistical and deep representation-learning approaches differ in their ability to rank and detect rare anomalies in high-dimensional hyperspectral imagery?

A secondary objective is to examine the difference between continuous anomaly-ranking performance and binary detection performance under severe class imbalance.

2 Dataset

2.1 AVIRIS-1 San Diego Scene

The experiments use the AVIRIS-1 San Diego hyperspectral dataset. The hyperspectral image contains an urban background with aircraft represented as anomalous objects.

The data are stored as a three-dimensional hyperspectral cube:

$$\mathbf{X} \in \mathbb{R}^{100 \times 100 \times 189}.$$

The corresponding binary ground-truth mask has dimensions

$$\mathbf{Y} \in \{0, 1\}^{100 \times 100},$$

where zero represents background and one represents an anomaly.

Table 1 summarizes the dataset.

Table 1: Characteristics of the AVIRIS-1 San Diego dataset used in this study.

Property	Value
Spatial dimensions	100×100 pixels
Spectral bands	189
Total pixels	10,000
Background pixels	9,936
Anomaly pixels	64
Anomaly proportion	0.64%
Anomalous objects	Aircraft

Only 64 of the 10,000 pixels are labeled as anomalies. Consequently, the dataset represents a highly imbalanced detection problem.

2.2 Data Quality

The hyperspectral cube was converted to 32-bit floating-point representation before analysis. The data-quality inspection examined minimum and maximum values, summary statistics, missing observations, infinite values, and the distribution of background and anomaly pixels.

No missing or infinite values were identified in the analyzed hyperspectral cube. The ground-truth map was converted to a binary mask for evaluation.

2.3 Spectral Representation

For pixel-level models, the hyperspectral cube was reshaped from

$$100 \times 100 \times 189$$

to

$$10,000 \times 189.$$

Each row therefore represents one spatial pixel and each column represents one spectral band.

This representation enables each pixel to be analyzed as a high-dimensional spectral observation.

3 Methodology

3.1 Experimental Design

The project was structured as a progression from a classical statistical detector to increasingly flexible representation-learning approaches.

The overall experimental workflow can be summarized as:

Hyperspectral Cube $\rightarrow$ Preprocessing $\rightarrow$ Feature Representation $\rightarrow$ Anomaly Scoring $\rightarrow$ Evaluation.

The ground-truth anomaly labels were used for evaluation rather than as conventional supervised training targets.

A fixed random seed of 42 was used throughout the experiments where stochastic operations were required.

3.2 Spectral Standardization

For the deep spectral models, the 189-dimensional pixel vectors were standardized using feature-wise normalization. For spectral band j , the standardized value is

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j},$$

where x_{ij} is the original value of spectral band j for pixel i , μ_j is the mean of the band, and σ_j is its standard deviation.

Standardization reduces differences in numerical scale between spectral bands and improves numerical stability during model training.

3.3 Regularized RX Detector

The Reed–Xiaoli detector was used as the statistical baseline. The RX detector measures the squared Mahalanobis distance between a pixel and the estimated background distribution.

For spectral feature vector $\mathbf{x}$,

$$RX(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}),$$

where $\boldsymbol{\mu}$ is the estimated background mean and $\boldsymbol{\Sigma}$ is the covariance matrix.

Because hyperspectral bands are strongly correlated, directly estimating a high-dimensional covariance matrix can be unstable. Principal Component Analysis (PCA) was therefore applied after standardization.

The original 189 spectral dimensions were reduced to 30 principal components while retaining approximately 99.98% of the observed variance.

A Ledoit–Wolf covariance estimator was then used to obtain a regularized covariance matrix. Larger RX scores indicate observations that are farther from the estimated background distribution.

For illustrative binary classification, pixels at or above the 99th percentile of the RX anomaly-score distribution were classified as anomalies. This percentile was treated as an operating threshold rather than an optimized decision boundary.

3.4 Dense Spectral Autoencoder

A dense autoencoder was developed to learn a compressed representation of the 189-dimensional spectral signatures.

An autoencoder contains an encoder

$$\mathbf{z} = f_{\theta}(\mathbf{x})$$

and a decoder

$$\hat{\mathbf{x}} = g_{\phi}(\mathbf{z}),$$

where $\mathbf{z}$ is a lower-dimensional latent representation.

The model was trained to minimize reconstruction error between the original spectrum and its reconstruction. For each pixel, reconstruction error was subsequently used as an anomaly score:

$$s(\mathbf{x}) = \frac{1}{B} \sum_{j=1}^B (x_j - \hat{x}_j)^2,$$

where $B = 189$ is the number of spectral bands.

The underlying assumption is that spectra consistent with the dominant data distribution should be reconstructed more accurately than unusual spectra.

The dataset contained 10,000 spectral observations. An 85%/15% training-validation partition was used during model development.

3.5 One-Dimensional Convolutional Autoencoder

The 1D convolutional autoencoder extends the dense reconstruction approach by explicitly processing neighboring spectral bands.

Each pixel was represented as

$$\mathbf{x} \in \mathbb{R}^{1 \times 189}.$$

One-dimensional convolutional filters operate along the spectral dimension, allowing the model to learn local relationships between neighboring wavelengths.

The encoder consisted of successive one-dimensional convolution, nonlinear activation, and pooling operations. The decoder reconstructed the spectral sequence using upsampling and convolutional layers.

As with the dense autoencoder, reconstruction error was used as the anomaly score.

3.6 Three-Dimensional Spectral-Spatial Autoencoder

Pixel-level models use spectral information but do not explicitly represent the spatial neighborhood surrounding each pixel. To investigate whether spatial context could improve anomaly detection, a three-dimensional spectral-spatial autoencoder was implemented.

The original 189-dimensional spectral representation was first standardized and reduced to 16 principal components. These components retained approximately 99.95% of the observed variance.

The reduced features were reshaped back into a spatial cube:

$$100 \times 100 \times 16.$$

A 9×9 spatial patch centered on each pixel was then extracted. The resulting representation contains both local spatial context and reduced spectral information.

A three-dimensional convolutional autoencoder was trained to reconstruct these local spectral-spatial structures. Center-pixel reconstruction error was used to generate anomaly scores.

This model tests whether local spatial context provides useful information beyond spectral signatures alone.

3.7 Spectral Variational Autoencoder

A variational autoencoder (VAE) was developed to introduce a probabilistic latent representation.

Unlike a deterministic autoencoder, the VAE represents each observation using parameters of a latent probability distribution. The encoder estimates a latent mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\sigma}^2$, from which latent representations are sampled.

The objective combines reconstruction loss with Kullback–Leibler divergence:

$$\mathcal{L}_{VAE} = \mathcal{L}_{reconstruction} + \beta D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z})).$$

The project examined reconstruction-based anomaly information, latent-space divergence, and a combined anomaly score.

Pixels that are difficult to reconstruct or inconsistent with the learned latent distribution receive larger anomaly scores.

3.8 PatchCore-Style Spectral Anomaly Detection

The final experiment investigated a PatchCore-style nearest-neighbor approach.

Compact spectral embeddings were extracted using the previously trained 1D convolutional autoencoder. Each of the 10,000 image pixels was represented by a 64-dimensional embedding.

A memory bank was constructed from high-confidence background representations. For a query embedding $\mathbf{z}$, the anomaly score was based on the distance to the nearest background representation:

$$s(\mathbf{z}) = \min_{\mathbf{m} \in \mathcal{M}} d(\mathbf{z}, \mathbf{m}),$$

where $\mathcal{M}$ represents the background memory bank.

Pixels whose learned representations are far from the background memory bank receive larger anomaly scores.

This approach differs fundamentally from reconstruction-based autoencoders because anomaly detection is performed in the learned representation space rather than using reconstruction error alone.

4 Evaluation Metrics

Because only 0.64% of the pixels are anomalous, overall accuracy is not an informative primary metric. A model predicting nearly all pixels as background could obtain very high accuracy while failing to detect the aircraft.

The experiments therefore emphasize several complementary metrics.

4.1 ROC-AUC

The area under the receiver operating characteristic curve evaluates how well anomaly scores rank positive observations above negative observations across thresholds.

4.2 Precision–Recall AUC

Precision–recall analysis is particularly important for rare-event detection because it focuses on the positive class.

Precision is defined as

$$\text{Precision} = \frac{TP}{TP + FP},$$

while recall is

$$\text{Recall} = \frac{TP}{TP + FN}.$$

4.3 F1-Score

The F1-score summarizes the balance between precision and recall:

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

4.4 Specificity and False-Positive Rate

Specificity measures the proportion of background observations correctly rejected:

$$\text{Specificity} = \frac{TN}{TN + FP}.$$

The false-positive rate is

$$FPR = \frac{FP}{FP + TN}.$$

These metrics are important in hyperspectral anomaly detection because even a small false-positive rate can produce many false detections when anomalies are extremely rare.

5 Results

5.1 RX Statistical Baseline

The regularized RX detector achieved strong ranking performance.

Table 2: Performance of the regularized RX baseline.

Metric	Result
ROC-AUC	0.9753
PR-AUC	0.1122
Precision	0.0100
Recall	0.0156
F1-score	0.0122
Specificity	0.9900
False-positive rate	0.009964

The ROC-AUC of 0.9753 indicates that the continuous RX scores generally ranked anomalous pixels above background observations.

However, the 99th-percentile threshold produced only one true-positive detection among the 64 anomaly pixels.

The corresponding confusion matrix was:

Table 3: RX confusion matrix at the evaluated 99th-percentile threshold.

	Predicted Background	Predicted Anomaly
Actual Background	9,837	99
Actual Anomaly	63	1

This result demonstrates an important distinction between ranking performance and binary detection performance. Although the RX score separates anomalies from background well across thresholds, the selected operating threshold provides poor anomaly recall.

5.2 Representation-Learning Models

Table 4 compares the representation-learning methods.

Table 4: Comparison of deep representation-learning anomaly detectors.

Model	ROC-AUC	PR-AUC	Precision	Recall	F1
Spectral Dense Autoencoder	0.9106	0.0346	0.0000	0.0000	0.0000
1D CNN Autoencoder	0.8944	0.0350	0.0000	0.0000	0.0000
3D Patch Autoencoder	0.9038	0.0423	0.0100	0.0156	0.0122
Spectral Variational Autoencoder	0.9259	0.0412	0.0000	0.0000	0.0000
Hyperspectral PatchCore-Style	0.9346	0.1441	0.2100	0.3281	0.2561

The results reveal substantial differences between ranking performance and threshold-based detection.

5.3 Dense Autoencoder

The dense spectral autoencoder achieved a ROC-AUC of 0.9106 and PR-AUC of 0.0346.

Although the ROC-AUC indicates that reconstruction error contained information related to anomalous behavior, the evaluated threshold produced no true-positive detections, resulting in zero precision, recall, and F1-score.

This indicates that reconstruction-error ranking alone was insufficient to produce useful binary detections under the selected threshold.

5.4 1D Convolutional Autoencoder

The 1D convolutional autoencoder achieved a ROC-AUC of 0.8944 and PR-AUC of 0.0350.

Despite explicitly modeling relationships between neighboring spectral bands, its ranking performance was slightly lower than the dense autoencoder. At the evaluated operating threshold, it also produced no true-positive detections.

This suggests that learning local spectral patterns did not by itself resolve the difficulty of separating rare aircraft anomalies from complex background spectra.

5.5 3D Spectral-Spatial Autoencoder

The 3D patch autoencoder achieved a ROC-AUC of 0.9038 and PR-AUC of 0.0423.

Its precision was 0.0100, recall was 0.0156, and F1-score was 0.0122.

The model therefore produced limited improvement in precision–recall ranking compared with the purely spectral autoencoders, but thresholded detection remained weak.

The result suggests that incorporating local spatial information changes the anomaly representation but does not automatically solve the false-alarm and threshold-selection problem.

5.6 Spectral Variational Autoencoder

The spectral VAE achieved a combined ROC-AUC of 0.9259 and PR-AUC of 0.0412.

The reconstruction component alone produced a ROC-AUC of 0.9109, while the KL-divergence component achieved a ROC-AUC of 0.9688.

The strong KL-divergence ROC-AUC suggests that latent-distribution information captured aspects of spectral abnormality that were not fully represented by reconstruction error.

However, the combined score did not translate into effective thresholded detection at the evaluated operating point, where precision, recall, and F1-score remained zero.

5.7 PatchCore-Style Detector

The PatchCore-style approach achieved:

Table 5: Performance of the PatchCore-style spectral anomaly detector.

Metric	Result
ROC-AUC	0.9346
PR-AUC	0.1441
Precision	0.2100
Recall	0.3281
F1-score	0.2561
Specificity	0.9920

Among the deep representation-learning methods evaluated in this project, the PatchCore-style approach produced the strongest precision–recall performance.

Its PR-AUC of 0.1441 was substantially higher than those of the dense autoencoder, 1D convolutional autoencoder, 3D patch autoencoder, and VAE.

At the evaluated threshold, the method achieved precision of 0.2100 and recall of 0.3281, producing an F1-score of 0.2561.

The result suggests that comparing learned representations against a memory bank of high-confidence background observations can provide a more useful anomaly score for rare-event detection than reconstruction error alone.

6 Discussion

6.1 Ranking Performance and Detection Performance Are Different

One of the clearest findings from this study is that a high ROC-AUC does not necessarily imply strong binary anomaly detection.

The RX detector achieved the highest observed ROC-AUC of 0.9753, indicating excellent ranking behavior. Nevertheless, at the evaluated 99th-percentile threshold, it detected only one of the 64 anomaly pixels.

Similarly, the VAE’s KL-divergence component achieved a ROC-AUC of 0.9688, yet the combined VAE score produced no true-positive detections at the evaluated operating threshold.

These results demonstrate why threshold-independent and threshold-dependent metrics should be interpreted together.

6.2 Importance of Precision–Recall Evaluation

The extreme imbalance of the AVIRIS scene makes precision–recall analysis particularly informative.

Only 0.64% of the pixels correspond to anomalies. Consequently, a detector can have a low false-positive rate and still generate substantially more false detections than true detections.

For this reason, ROC-AUC alone may give an incomplete view of practical anomaly-detection performance.

The PatchCore-style model illustrates this distinction. Its ROC-AUC of 0.9346 was lower than the RX detector’s 0.9753, but its PR-AUC of 0.1441 exceeded the RX baseline PR-AUC of 0.1122 and substantially exceeded the reconstruction-based deep models.

It also produced meaningfully higher thresholded precision and recall.

6.3 Reconstruction Error Has Limitations

The dense, 1D convolutional, 3D, and variational autoencoder experiments demonstrate that successful reconstruction learning does not necessarily guarantee strong anomaly separation.

Autoencoders are trained to minimize reconstruction error over the data distribution. When anomalous observations are present during unsupervised training, sufficiently expressive models may also learn to reconstruct them.

Furthermore, reconstruction-error distributions for background and anomaly observations may overlap substantially.

This can explain why several autoencoder models achieved ROC-AUC values around 0.90 while performing poorly after thresholding.

6.4 Spectral-Spatial Context

The 3D autoencoder incorporated spatial neighborhoods using 9×9 patches and 16 PCA components.

The model achieved a slightly higher PR-AUC than the dense and 1D spectral autoencoders, indicating that local spatial context contained some useful anomaly information.

However, the improvement was not sufficient to produce strong binary detection.

Future spectral-spatial architectures could investigate alternative patch sizes, more targeted background-only training, attention mechanisms, or architectures designed specifically for hyperspectral representation learning.

6.5 Representation-Space Distance

The PatchCore-style detector produced the strongest practical detection performance among the deep models.

Instead of assuming that anomalies necessarily reconstruct poorly, this approach asks whether an observation’s learned representation resembles known high-confidence background representations.

This distinction appears useful for the AVIRIS scene.

The result suggests that representation-space distance may provide a useful complement to reconstruction-based anomaly scores in severely imbalanced hyperspectral detection problems.

7 Limitations

Several limitations should be considered when interpreting the results.

First, all experiments were conducted using a single AVIRIS hyperspectral scene. Performance on this scene should therefore not be interpreted as evidence of generalization to other sensors, geographic regions, environmental conditions, or anomaly types.

Second, the ground-truth anomalies correspond to aircraft in an urban scene. Other hyperspectral anomaly-detection problems may involve substantially different spectral and spatial characteristics.

Third, threshold-based metrics depend strongly on the selected operating threshold. The percentile thresholds used in these experiments provide a consistent experimental mechanism but were not optimized for operational deployment.

Fourth, several unsupervised models were trained using data that contain both background and anomaly observations. Because anomalies are rare, the dataset is dominated by background pixels,

but contamination of the training distribution remains possible.

Fifth, the experiments compare model families but do not constitute an exhaustive hyperparameter search. Alternative architectures, latent dimensions, training strategies, patch sizes, and anomaly-score combinations may produce different results.

Sixth, the evaluation is performed at the pixel level. Aircraft occupy spatially connected regions, meaning that future work could additionally evaluate object-level detection performance.

Finally, the results represent an independent computational investigation rather than a benchmark study across multiple established hyperspectral anomaly-detection datasets.

8 Future Work

Several extensions could strengthen the analysis.

1. **Multi-scene validation.** Evaluate the models on additional hyperspectral anomaly-detection datasets to determine whether the observed behavior generalizes beyond the AVIRIS-1 San Diego scene.
2. **Background-only training.** Investigate strategies for constructing cleaner background training sets and reducing contamination by anomalous observations.
3. **Threshold optimization.** Compare percentile thresholds with methods designed to control false-alarm rates or select operating points using validation data.
4. **Local statistical detectors.** Compare the global RX detector with Local RX, Kernel RX, and robust covariance approaches.
5. **Spectral-spatial representation learning.** Investigate more advanced architectures that jointly model spectral and spatial structure.
6. **Ablation analysis.** Study the influence of PCA dimensionality, latent-space dimension, patch size, memory-bank construction, and anomaly-score formulation.
7. **Multiple random seeds.** Repeat neural-network experiments across several random seeds and report mean performance and uncertainty.
8. **Object-level evaluation.** Complement pixel-level metrics with connected-component or object-level aircraft detection measures.
9. **Cross-domain evaluation.** Evaluate whether representations learned from one hyperspectral scene transfer to independent scenes or sensors.

9 Conclusion

This independent research study investigated statistical and deep representation-learning approaches for unsupervised hyperspectral anomaly detection using the AVIRIS-1 San Diego scene.

Six approaches were examined: regularized RX detection, a dense spectral autoencoder, a 1D convolutional autoencoder, a 3D spectral-spatial autoencoder, a spectral variational autoencoder, and a PatchCore-style representation-distance detector.

The experiments reveal three primary findings.

First, the regularized RX detector provided very strong anomaly-ranking performance, achieving a ROC-AUC of 0.9753. However, its thresholded detection performance was weak, demonstrating that high ROC-AUC does not necessarily imply effective rare-event detection.

Second, reconstruction-based neural models captured anomaly-related information but generally struggled to separate rare anomalies at the evaluated operating thresholds. The dense autoencoder, 1D convolutional autoencoder, 3D spectral-spatial autoencoder, and VAE all produced ROC-AUC values near or above 0.89, while their threshold-based F1-scores remained low.

Third, the PatchCore-style detector produced the strongest practical detection performance among the evaluated deep representation-learning methods. It achieved a ROC-AUC of 0.9346, PR-AUC of 0.1441, precision of 0.2100, recall of 0.3281, and F1-score of 0.2561.

Overall, the study demonstrates the importance of evaluating hyperspectral anomaly-detection systems using multiple complementary metrics. In severely imbalanced remote-sensing problems, ranking metrics, precision–recall behavior, threshold sensitivity, false-alarm rates, and spatial detection patterns provide different and complementary information about model performance.

The results also motivate further investigation of representation-learning methods that combine spectral information, spatial context, and background-reference modeling for robust hyperspectral anomaly detection.

Reproducibility

The experiments were implemented in Python using NumPy, SciPy, Pandas, Scikit-learn, PyTorch, Matplotlib, Spectral Python, and Joblib.

A fixed random seed of 42 was used where applicable. Preprocessing objects, trained models, anomaly scores, learned embeddings, and model-comparison tables were saved during the experimental workflow to support reproducibility.

The project code and notebooks are maintained in the author’s GitHub repository.

Independent Research Statement

This report documents an independent computational research project conducted by Monica Chaganti. The work was undertaken to investigate machine learning and deep representation-learning methods for remote-sensing anomaly detection and to develop experience in reproducible experimentation with high-dimensional hyperspectral imagery.

This document is an independent research report and is not presented as a peer-reviewed publication.

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